Market Lens

Release market intelligence for Indian music labels - detailed project report

Prepared 23 October 2026 | Data: songs first uploaded 2026-09-15 to 2026-10-15; 3 days of daily readings | Status: working pilot on a development laptop

6 labels	**76** official channels	**1,436** songs in window	**1.34B** views on those songs
14,028 YouTube videos collected	**97%** songs with singer credits	**466** songs matched on Spotify	**524** songs with ISRC

Executive summary

Market Lens collects public data about new releases from the official YouTube channels of six major Indian labels (Sony Music India, T-Series, Zee Music Company, Saregama, Aditya Music and Think Music). It groups every upload of a song into one record, enriches it with metadata from Spotify, Apple Music, MusicBrainz, Deezer and the credits in the video descriptions, and records views, likes and comments every 6 hours. Matching, merging and every metric follow fixed, explainable rules. A web console (Market Lens Console) runs the pipeline and lets the team review, explore and export. An AI Insights page ranks emerging hits, compares labels fairly and finds catalogue opportunities; a written AI-assist assessment explains the numbers, and a live LLM layer is built but switched off pending approval.

- **Coverage:** credits and language for 97-100% of songs; Spotify metadata for about a third, limited by Spotify's development-mode quotas, with Apple Music filling another 43%.
- **Headline insight:** volume is not performance - T-Series publishes the most songs and views but its typical song ranks below Think Music, Saregama and Sony against songs of the same age.
- **Main constraints:** Spotify's daily quotas and lack of play counts; no historical data before collection starts; terms-of-service sign-off needed before commercial use.
- **Most urgent next step:** deploy to the server as a service so daily history accumulates without gaps.

1. Background and scope

Times Music agreed a Phase 0 feasibility exercise with Saranyu for a release market-data platform. Market Lens is the working pilot for that exercise.

Phase 0 target	Status in the pilot
Discover >= 95% of new releases within 24 hours	All uploads on the tracked channels are collected every 6 hours
Match >= 90% across platforms	Rule-based matching to Spotify, Apple Music, MusicBrainz and Deezer; Spotify limited by quota
Capture >= 99% of daily readings	Readings every 6 hours in a video's first week, then daily; depends on the collector running
<= 10% of matches needing manual review	Uncertain matches go to Review; queues cleared after each run
Power BI-ready output	Per-run CSVs (songs, videos, metadata); daily totals tables planned in PostgreSQL
No AI/ML in matching	All matching, merging and metrics are rule-based; AI only explains (switched off live)

Decisions taken along the way

- Start with Sony Music India on Spotify (Phase 1), then switch to YouTube-first because Spotify's development mode gives no popularity and small quotas.
- Song window set to songs first uploaded 15 September to 15 October 2026 (configurable); uploads collected from 31 days earlier.
- Extend to six labels using each label's own featured channels; no scraping of any platform.
- Keep live AI off for now; show a written AI-assist assessment instead and refer to the integration as 'LLM'.

2. What was implemented, in order

#	Milestone	What it delivered
1	Spotify discovery (Sony India)	Label search, album and track metadata, ISRC merge, quota-aware incremental runs
2	Console	Web UI to configure, run, review and download; password protected
3	YouTube connector	8 Sony channels, uploads, views/likes/comments, video-to-track matching, Review page
4	YouTube-first songs	Songs grouped from uploads; one Spotify search per song; Deezer rank by ISRC
5	Review and decision log	Approve/reject/link with who and when; exact undo
6	Multiple YouTube keys	11 keys used in order; quota tracked per key
7	Six labels	Aditya, Saregama, Think, T-Series, Zee added; 76 channels; labels can be switched on/off
8	Song versions and merging	Per-type stats, Spotify versions, four merge rules, bulk merge, aliases
9	Song metadata and Explore	Credits parser, language/category/occasion rules, metrics per metadata value
10	More metadata sources	Artist-first Spotify lookups, Apple iTunes, MusicBrainz
11	AI Insights	Five rule-based panels, song benchmarks, written AI-assist briefing; LLM layer built and off
12	Data coverage view	Live coverage and 'not available from the APIs' on the Sources page

3. Architecture and pipeline

One pipeline runs every 6 hours inside the Console process (and on demand). Each step is independent: a failing or quota-blocked step does not stop the next one, and every step resumes where it stopped.

Step	What happens	Rules and safeguards
1. Collect uploads	Each channel's uploads since the window start minus 31 days	Incremental: stops at known or older videos; 1 quota unit per 50 videos
2. Read views	Views, likes, comments, tags, audio language for due videos	Every run in a video's first 7 days, daily after; every reading kept
3. Group into songs	Uploads with the same song name (per label) form one song	Remix, male/female and 'X version' stay separate; shorts/teasers only join
4. Spotify search	One search per song for ISRC, album, release date, artists	Scored match; uncertain to Review; misses retried every 3 days
5. Spotify artist-first	Credited artists' recent singles and albums	Works while search is blocked; confident matches only
6. Apple iTunes	Album, release date, artists, genre for songs Spotify hasn't reached	Confident matches only; 3 s between calls
7. MusicBrainz	ISRC for songs without one	1 request a second; confident matches only
8. Merge rules	Duplicate songs merged or suggested	Configurable; undoable; aliases keep merges
9. Metadata	Credits, film, language, category, occasion per song	Rebuilt every run; source recorded per value
10. Deezer rank	Daily rank per ISRC	Free API; search not usable from India
11. Outputs	Songs CSV, videos CSV, metadata CSV	Raw API responses stored for audit and replay

Technology

- Python 3 standard library (no external dependencies for the pipeline), SQLite in WAL mode, single-page web console.
- Quota handling: YouTube keys rotate on quota exhaustion; Spotify stops cleanly on a daily block and the health watcher restarts the pipeline when access returns.
- Scale today: 14,028 videos, 6,339 stored raw API responses, 27 runs logged.

4. Labels and channels

Label	Channels	Songs	Channels included
T-Series	25	619	Main; Telugu, Tamil, Kannada, Malayalam, Marathi, Gujarati, Bangla, Apna Punjab, Hamaar Bhojpuri, Haryanvi, Rajasthani, Himachali, Kannada folk, Regional; Bhakti Sagar (5 languages), Bhakti Marathi, Prarthana, Shabad Gurbani, Islamic, Classical
Saregama	19	280	Main; Tamil, Telugu, Malayalam, Kannada, Bengali, Marathi, Gujarati, Punjabi, Hum Bhojpuri, Odia, Chhattisgarhi, Mharo Rajasthan; Bhakti, South Devotional, Ghazal, Sufi, Carnatic, Hindustani
Sony Music India	8	204	Main, VEVO, South, South VEVO, Malayalam, Bangla, Bhakti, Classical
Aditya Music	7	163	Main, Hindi, Tamil, Kannada, Malayalam, Bhakthi, Bhakthi Hindi
Zee Music Company	12	140	Main, South, Marathi, Bangla, Bhojpuri, Haryanvi, Chhattisgarhi, Gujarati, Odia, Rajasthani, Devotional, Ibadat
Think Music	5	27	India, Telugu, Malayalam, Think Indie Collective, Think Divine

Selection method: the channels each label's main channel features on its own page (Zee features none, so its channels were verified by contact details and audience size). Left out: movie, TV-show, kids, karaoke, remix/jhankar, old-song re-upload and look-alike channels (about 40 for Zee alone).

5. Songs and versions

A label usually publishes three or four versions of a song. Market Lens keeps every upload under one song with its own views, likes and comments, and totals by type.

Version type	How it is recognised	Views (window songs)
Video song	Full-length upload that is not lyric, audio, short or teaser (titles often omit 'video')	444M
Lyric video	'Lyric' in the title	128M
Teaser / promo	Teaser, promo, glimpse, trailer, sneak peek, first look	19M
Audio	'Audio', visualizer	11M
Shorts	75 seconds or less, or #shorts	9M
Jukebox (excluded)	Jukebox, mashup, non-stop, full album, over 15 minutes	not a song

- Shorts and teasers join an existing song but never create one (their captions are often not the song name).
- Spotify: the single, album and re-release copies of the same recording are linked to one song (same ISRC, or same title, main artist and length within 3 s), at no extra API cost.
- Songs never merge across labels; song names in Indian scripts keep their vowel signs.

6. Song metadata

Labels publish structured credits in their video descriptions ('Singer : X', 'Music - Y', 'Lyrics: Z', 'Directed by W'). A rule-based parser reads them, ignoring promotional lines and lyrics, and every value records its source.

Field	Source	Coverage
Language	Title words, YouTube audio language, channel, title script, tags (in that order)	100%
Singers	YouTube credits; Spotify or Apple artists when credits have none	97%
Music director	YouTube credits	97%
Lyricist	YouTube credits	87%
Release date	Spotify or Apple Music	75%
Film	Credits, Spotify/Apple soundtrack titles, 'X Movie' in titles	49%
Genre	Apple Music	43%
Cast	YouTube credits	40%
Category, occasion	Channel and keyword rules	100%

Categories and occasions (window songs)

Category	Songs
Film	655
Non-film / Indie	452
Devotional	255
TV / Web show	60
Classical	9
Ghazal / Sufi	5

Occasions detected: Navratri 101, Ganesh Chaturthi 74, Durga Puja 38, Dussehra 6, Janmashtami 3, Diwali 2, Chhath 2, Holi 1, Birth anniversary / tribute 1.

Most frequent credits

Singer	Songs	Music director	Songs
S.P. Balasubrahmanyam	24	GV Prakash Kumar	42

Singer	Songs	Music director	Songs
Arijit Singh	19	Shamir Tandon	38
Karthik	19	A.R.Rahman	27
Shreya Ghoshal	17	Traditional	27
Shilpi Raj	16	Thaman S	25
K.S. Chithra	15	Vasuki Vaibhav	25
Sumanth Prabhas	14	Devi Sri Prasad	23
Mani Aegurla	13	Jakes Bejoy	23
Narendra Ravi	13	Mani Sharma	22
Udit Narayan	13	Anirudh Ravichander	21

7. Matching, merging and review

Rule	Detail
Match score	Title 55%, artists 15%, film/album 10%, length 10%, date 10%
Automatic match	Total >= 0.75 and title >= 0.85 (or title >= 0.95 with artist confirmation); at least one corroborating signal
Sent to Review	Total >= 0.55; never a remix, male/female, instrumental or 'X version' against the original
Merge rule: same ISRC	Merges automatically
Merge rule: same Spotify title + artist	Suggests; within 60 days and 5 s length
Merge rule: spelling variant	Suggests; name similarity >= 0.9, within 30 days and 8 s
Merge rule: name contains the other	Suggests; at least 2 words, within 14 days and 8 s
Undo	Every decision stores a before-snapshot and can be undone exactly; merged names become aliases

Review work so far: 21 approve, 1 merge, 6 reject decisions recorded and active.

Bugs found through review and fixed

- Indian-script vowel signs were dropped from song names.
- Merges failed for songs that already had metadata.
- Generic phrases ('Durga Puja Original Song') were taken as song names.
- A renamed song lost its Spotify link; songs now keep their identity when names change.
- 'Full Album' uploads were counted as songs; ordinary words ('Album') counted as artist evidence.

8. Cross-platform lookups

Source	Method	Result to date
Spotify search	One search per song, scored	466 songs matched; 194 not on Spotify
Spotify artist-first	Credited artists' recent releases, then tracks	19 new releases matched before the album quota ran out
Apple iTunes	Search by name + singer, scored	622 songs with album, date, genre
MusicBrainz	Recording search by name + singer, scored	170 recordings matched; ISRC on 524 songs overall
Deezer	Track by ISRC	Rank for 514 songs

9. The Market Lens Console

Page	What the team can do
Overview	KPIs, songs per day, songs by label and channel, top songs by views, biggest day-on-day gainers
Songs	Search (including credits), filter by label, channel, language, category, status and date range; play on YouTube or Spotify; select songs to merge; export CSV
Song panel	Release details and credits with sources, comparison with similar songs, views by version type, trend chart, every video and Spotify version
Explore	Rank singers, composers, lyricists, cast, films, languages, categories, occasions, labels, channels and genres by songs, views, gain, engagement; drill into any value
AI Insights	Five insight panels with written AI-assist notes; data-history notice; Ask Music AI marked as coming with the LLM integration
Review	Spotify links to check, duplicate songs (merge all of a rule in one click), video matches, decision history with undo
Runs / Outputs	Start or cancel runs, live log, run history; preview and download every CSV and raw API response
Sources	API keys and quota per key, LLM key (planned), data coverage and gaps, all data sources
Settings	Release window, labels and channels, merge rules, schedule, health watcher

10. Data coverage

Data	Source	Songs	Coverage
YouTube views	youtube	1,434	100%
Spotify match (metadata)	spotify	466	32%
Spotify lookup still pending	spotify	776	54%
Not on Spotify (searched)	spotify	194	14%
Apple Music metadata	apple	622	43%
ISRC	spotify / musicbrainz	524	36%
Deezer rank	deezer	514	36%
Release date	spotify / apple	1,070	75%
Singers	credits	1,400	97%
Music director	credits	1,399	97%
Lyricist	credits	1,246	87%
Film	credits / spotify / apple	704	49%
Cast	credits	575	40%
Language	rules	1,434	100%
Genre	apple	622	43%

The Spotify gap is a daily quota, not missing data: about 600 searches a day clear the backlog in 1.5-2 days.

11. Insights

Method. Every song is compared with songs of the same age band (0-7, 8-30, 31-90, 90+ days since first upload), so older uploads and bigger labels don't win by default. 'Percentile' = share of same-age songs a song beats on views per day (50 = typical). Daily readings exist for 3 day(s) (2026-10-16, 2026-10-22, 2026-10-23). Growth, acceleration and anomaly signals become reliable after 4-7 days of daily readings; until then rankings lean on age-adjusted views per day.

11.1 Labels compared fairly

Label	Songs	Median percentile	Above typical	Likes per 1k	Total views
Think Music	27	85	96%	4.1	81.9M
Saregama	281	57	58%	6.3	320.2M
Sony Music India	204	56	56%	6.1	183.1M
Aditya Music	163	48	47%	4.8	66.8M
Zee Music Company	140	46	44%	6.2	148.7M
T-Series	619	44	45%	6.3	578.1M

T-Series publishes the most songs and views, but many uploads are small regional and devotional releases, which pulls its typical song below the others. Think Music is small but consistent.

11.2 Languages

Language	Songs	Median percentile	Above typical	Likes per 1k
Rajasthani	6	86	100%	10.8
English	26	66	58%	8.4
Hindi	360	66	76%	7.0
Chhattisgarhi	7	63	71%	7.2
Haryanvi	19	63	53%	10.7
Bhojpuri	81	61	65%	6.3
Bengali	44	46	43%	9.2
Tamil	165	43	47%	5.1
Telugu	394	39	37%	4.4
Marathi	57	39	35%	5.6
Odia	6	38	50%	12.8
Malayalam	97	36	35%	5.0
Punjabi	90	32	31%	8.0
Gujarati	44	28	32%	9.9
Kannada	36	20	39%	4.5

11.3 Strongest and weakest label x language pockets (5+ songs)

Strongest	Pct	Weakest	Pct
T-Series - Haryanvi (10)	91	T-Series - Marathi (30)	14
T-Series - Rajasthani (5)	88	T-Series - Tamil (42)	14
Think Music - Telugu (7)	87	Zee Music Company - Tamil (8)	18
Think Music - Tamil (16)	85	T-Series - Malayalam (14)	19
Zee Music Company - Gujarati (7)	85	Saregama - Punjabi (15)	19
T-Series - Bengali (15)	84	Zee Music Company - Malayalam (22)	20
Saregama - Kannada (10)	78	T-Series - Gujarati (23)	22

Strongest	Pct	Weakest	Pct
Saregama - Chhattisgarhi (5)	72	Saregama - Gujarati (11)	25

11.4 Emerging hits (not yet in the top 5% by views)

Score	Song	Label / language	Age (days)	Views	Views per day	Likes per 1k
98	Tulsi Kumar	T-Series / Hindi	10	3.4M	335,682	11.4
97	Awadhiya Ke Raja - From "Nayyi Navelli"	Zee Music Company / Hindi	9	3.4M	388,006	0.7
95	Ramuloo Ramulaa	Aditya Music / Telugu	8	1.0M	133,993	4.5
95	Ambabai Majhi Tu Maay	Saregama / Marathi	13	1.7M	126,899	4.9
95	Aashiq Banaya Aapne	T-Series / Hindi	27	3.7M	137,858	3.5
95	Jim Jim Tara (From "Parnashala")	Aditya Music / Telugu	10	2.1M	208,782	7.4
95	Tiger Siva	Saregama / Tamil	17	3.5M	209,774	5.8
94	Ek Radha Ek Meera	Saregama / Hindi	22	2.4M	106,999	3.8
94	Nadha Kalipothundi (From "Garividi Laksh	T-Series / Telugu	13	1.9M	150,195	4.1
93	Inkem Inkem Kavale	Aditya Music / Telugu	10	760K	78,659	3.9
93	Aigiri Nandini	Zee Music Company / Hindi	13	1.7M	131,755	5.2
93	Kurchi Madathapetti	Aditya Music / Telugu	31	2.5M	82,280	4.4
93	Navratri Special	T-Series / Marathi	15	2.4M	160,217	9.4
93	Dhumavati	Saregama / Telugu	8	1.3M	166,131	9.0
92	If You Come Today 2.0	Saregama / Kannada	15	913K	61,876	1.0

Most scores are early signals based on age-adjusted speed; growth and acceleration sharpen with more daily readings. Fast songs with very few likes per 1,000 views (typical is about 6) may reflect promoted views - a hypothesis public data can't confirm.

11.5 Catalogue opportunities (released a year or more before the upload)

Song	Label	First released	Upload	Views per day	Percentile
Virisina (From "Raajahyogam")(Telugu)	Aditya Music	2022-12-16	2026-10-06	606,974	98
Nagada Sang Dhol	Sony Music India	2013-04-01	2026-10-09	406,458	97
Lallati Bhandar	Sony Music India	2013-07-30	2026-10-08	184,449	94
Kurchi Madathapetti	Aditya Music	2023-12-30	2026-09-22	82,280	92
Nit Khair Manga	T-Series	2018-03-01	2026-10-15	137,767	92
Ramuloo Ramulaa	Aditya Music	2019-09-28	2026-10-15	133,993	92
YAAD HAI NA	T-Series	2016-06-07	2026-09-23	115,487	91
Desi Look	T-Series	2015-02-20	2026-09-20	46,374	88
Bappa	Sony Music India	2016-08-23	2026-09-16	45,095	88
Achha Sila Diya	T-Series	2023-01-19	2026-10-07	94,663	89
Aakasam Enatido	Aditya Music	1982-01-01	2026-10-06	85,444	89
Inkem Inkem Kavale	Aditya Music	2018-07-10	2026-10-13	78,659	88

Festival favourites re-released in the festival weeks (Navratri, Ganesh Chaturthi, Durga Puja) are among the fastest uploads.

11.6 Artist portfolio health - singers (3+ songs)

Artist	Songs	Profile	Median pct	Top song share	Total views	Top song
Emiway Bantai	3	Mixed	99	63%	50.8M	Prahaar
NIJJAR	5	Consistent	98	49%	81.9M	Bari NIJJAR
Nishant Mukta	3	Mixed	98	67%	22.0M	UP Wali Laiki
V.M. Mahalingam	3	Mixed	97	69%	16.9M	Kobali (From "Ranabaali")
Hiphop Tamizha	4	Consistent	96	42%	26.8M	Vibe Venuma
Sai Abhyankkar	4	One-hit dependent	95	75%	52.7M	Ooroda Oththa Don
Vishal Mishra	11	Mixed	94	50%	63.8M	Prahaar
Ajay Gogavale	3	One-hit dependent	94	95%	56.0M	Endhayya Saami
Shivani Singh	5	Mixed	91	68%	31.3M	Pawan Singh
Anmol Salh	3	Consistent	90	47%	3.9M	Judaai
Siddharth Basrur	3	One-hit dependent	90	94%	20.1M	Vingansa
Swara Verma	3	Consistent	90	50%	6.8M	Those Pe Badmashi

11.7 Artist portfolio health - music directors (3+ songs)

Artist	Songs	Profile	Median pct	Top song share	Total views	Top song
Ajay	6	Mixed	98	53%	48.1M	Ranabaali
Atul	6	Mixed	98	53%	48.1M	Ranabaali
NIJJAR	5	Consistent	98	49%	81.9M	Bari NIJJAR
Karma	3	Mixed	98	50%	37.7M	Vingansa
Nishant Mukta	3	Mixed	98	67%	22.0M	UP Wali Laiki
Ravi Basrur	4	Consistent	97	42%	45.0M	Vingansa
Hiphop Tamizha	4	Consistent	96	42%	26.8M	Vibe Venuma
Sai Abhyankkar	4	One-hit dependent	95	75%	52.7M	Ooroda Oththa Don
Rochak Kohli	13	Consistent	93	25%	81.9M	Purane Din (From "Pehla Pyaar
Vishal Mishra	7	Mixed	92	67%	48.0M	Prahaar
Anmol Salh	3	Consistent	90	47%	3.9M	Judaai
Charan Arjun	3	Consistent	89	43%	5.2M	Ghal Ghal Gajjala Motha Garivi

'Consistent' = most songs above typical and no song over half of the views; 'One-hit dependent' = one song brings 70%+ of views. Known data issue: duo names written 'Ajay - Atul' also appear as 'Ajay' and 'Atul'.

11.8 Anomaly watch

8 songs currently show a sharp change in growth, mostly Sony titles, because only Sony was tracked on 16 October. Early 'spikes' compare uneven reading gaps and 'drops' are normal post-launch slowdowns; this panel becomes meaningful after about a week of daily readings across all labels.

12. AI layer

Part	Status	Detail
Insight panels and benchmarks	Live	Computed by rules; refresh with every run
AI-assist briefing and notes	Live (static)	Written assessment of the panels, dated 23 Oct; does not update by itself
Live LLM explanations	Built, off	One cached explanation per panel per day, using only the panel's numbers
Ask Music AI	Built, off	Chat with read-only tools: find songs, insight panels, group metrics, song details, read-only SQL

Part	Status	Detail
Safeguards	Built	No invented numbers; causes only as hypotheses; read-only database access; refusal fallback

Switching the live LLM on needs an LLM key on the Sources page and two endpoints enabled, after approval.

13. Outputs and data model

Output / table	Contents
market_lens_songs__run.csv	One row per song: dates, status, ISRC, label, channel, metadata columns, views/likes/comments, per-type views, IDs and links
market_lens_song_metadata__run.csv	One row per metadata value (song, field, value, source) - Power BI friendly
market_lens_youtube__run.csv	One row per YouTube video with its latest metrics and match
track / platform_item	Songs; every YouTube video, Spotify, Apple, MusicBrainz and Deezer item linked to a song
reading / snapshot	Every reading; the latest value per item per day
song_meta / song_alias	Metadata values with sources; merged names
review_decision / merge_rejection	Decision log with before-snapshots; pairs marked 'not the same'
raw_capture / run_log / api_check	Every raw API response; run history; API health checks

14. API limitations

Source	Limit	What it does not give	Handling
YouTube Data API	10,000 units/day per project (resets 12:30 PM IST); 1 unit per 50 videos; search 100 units	Watch time, impressions, retention, revenue, demographics, dislikes, past daily views	11 keys in order; ~300-500 units per daily sync; collect daily
Spotify (development mode)	~600 searches and ~90 album lookups a day; 10 results per page; batch calls blocked	Play counts (any mode), popularity, label, genres	Clean stop and auto-resume; artist-first route; extended quota application
Apple iTunes Search	~20 requests a minute	ISRC, plays	3 s spacing; ~50 min per 1,000 songs
MusicBrainz	1 request a second; User-Agent required	Most new Indian releases; popularity	Retried weekly
Deezer	50 requests / 5 s	Search from India; stream counts	Rank by ISRC only
LLM (planned)	Per-organisation rate tiers; billed per token	-	Daily cache per panel

Terms of service: Spotify prohibits scraping and restricts commercial analytics and derived metrics; YouTube limits how long API data may be stored without refresh; Apple, Deezer and MusicBrainz have their own terms. Scraping was not used. A terms-of-service sign-off is needed before commercial use.

15. Database and infrastructure

Estimate (all major Indian labels, from Jan 2025)	Value
Videos, Jan 2025 to end 2026	about 300,000-500,000, then about 250,000 a year
Songs	about 100,000-150,000, then about 70,000 a year
Daily reading rows	about 180M a year (1.5-2B over 5 years)
Readings storage	about 20 GB a year, 1-2 GB compressed
Raw API responses	about 125 GB a year: keep in object storage

Part	Recommendation
Main database	PostgreSQL 16 - open source (PostgreSQL License), free for commercial use
Time series	TimescaleDB - compression and automatic daily totals; free self-hosted (source-available licence; not resaleable as a service)
Raw responses	S3 (or MinIO / disk), compressed files per day
Reporting	Power BI on daily total tables (or Apache Superset / Metabase as open-source options)
Server	4 vCPU, 16-32 GB RAM, 200 GB SSD for the first 2-3 years; ClickHouse only at tens of billions of rows

16. Security and operations

- The Console is password protected (one shared password) and listens on port 3000; API secrets stay in a server-side .env and are never sent to the browser.
- API keys were shared in chat during the pilot and should be rotated.
- The collector currently runs as a background task on a development laptop: it stops if the session closes or the laptop sleeps. Production needs it as a service on the server.
- Every run, decision and raw response is logged for audit.

17. Risks and next steps

#	Next step	Why it matters
1	Deploy to the server and run as a service	Daily history only builds while collection runs every day
2	Spotify extended quota (as a company)	Development-mode limits block scale
3	Terms-of-service sign-off for every source	Required before commercial use
4	PostgreSQL + TimescaleDB; Power BI on daily totals	Scale, concurrency, direct reporting
5	Name clean-up (duo names, singer-as-title, spelling variants)	Cleaner artist rankings
6	Refresh insights after 7+ days of readings	Growth and anomaly signals become reliable
7	Switch on the LLM after approval	Live explanations and Ask Music AI
8	Optional: Spotify Charts CSV upload, Last.fm	Real Spotify streams for the top 200; listener counts
9	Per-user logins, HTTPS, key rotation	Security for a shared tool

Glossary

Term	Meaning
Song window	Songs first uploaded between the configured dates (15 Sep - 15 Oct 2026)
Views per day (velocity)	A song's views across all versions divided by days since its first upload
Percentile	Share of songs of the same age band that a song beats on views per day (50 = typical)
Breakout score	0-100 blend of percentile speed, growth relative to size and acceleration
New release / older release	On Spotify at most 30 days before the first YouTube upload / earlier or only on a compilation

Term	Meaning
ISRC	International Standard Recording Code: the global ID of a recording, used to join platforms

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